Relative Covariance

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Note: *Parts of this document may be AI-written. My general workflow is to write sections of the article myself, laying out the general points I want to hit on and going into detail where relevant. I then converse with an LLM to identify errors in the logic or math and opportunities to use clearer or more generalizable notation. I then have the LLM make the edits to reflect our proposed changes, including adding more detailed derivations.*

This article describes how the phenotypic covariance between two individuals i and j can be modeled by limited information we have about the pair. I derive formulas from the starting trait generative model and only introduce assumptions at the point the derivation requires it. I start with a simple trait architecture and iteratively add more complexity.

1 Additive genetics under random mating with independent environment

We begin with the simplest architecture: a trait determined by an additive genetic component and an independent environmental component, in a randomly mating population.

1.1 The phenotype model

For an individual i we write the (standardized) phenotype as

$$y_i = g_i + e_i, \tag{1}$$

where g_i is the genetic value and e_i the environmental deviation. We take y to be standardized across the population, $\mathrm{E}[y] = 0$ and $\mathrm{Var}[y] = 1$, and both components to be mean-centered, $\mathrm{E}[g] = \mathrm{E}[e] = 0$, with variances $\mathrm{Var}[g] = V_A$ and $\mathrm{Var}[e] = V_E$.

We then introduce the following assumptions.

[Additivity] Additive genetic value

The genetic value is a fixed additive function of the individual's own genotype,

$$g_i = \sum_{k=1}^{M} \beta_k \, x_{ik},$$

where β_k is the per-standardized-allele effect of variant k, x_{ik} is the standardized genotype ($\mathrm{E}[x_{ik}] = 0$, $\mathrm{Var}[x_{ik}] = 1$), and M is the number of variants in the genome. Non-causal variants are included with $\beta_k = 0$.

Commonly bundled into: the additive model; no dominance, no gene–gene interactions (GxG, epistasis), and no gene–environment interactions (GxE) — each would make the genetic value depend on more than the individual's own genotype in an additive way.

[GE-indep] Gene–environment independence

An individual's environmental component is independent of their own genetic component:

$$g_i \perp e_i$$

Commonly bundled into: no gene–environment correlation or interaction, within an individual.

[No-IGE] No indirect genetic effects

One individual's genetic component is independent of another individual's environmental component:

$$g_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no indirect (e.g. parental) genetic effects — one individual's genotype does not shape another's environment.

[Env-indep] Independent environments

The environmental component is independent across individuals, including relatives:

$$e_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no shared/common environment (C); no environmental confounder shared by the pair (e.g. population stratification).

Because $g_i \perp e_i$ (**[GE-indep]**), $\mathrm{Cov}[g_i, e_i] = 0$ and the phenotypic variance decomposes as $\mathrm{Var}[y] = V_A + V_E$. Narrow–sense heritability is typically defined as:

$$h^2 = \frac{V_A}{\mathrm{Var}[y]}$$

Since $\mathrm{Var}[y] = 1$, $h^2 = V_A$ under this model.

1.2 Phenotypic covariance is genotypic covariance

For a single fixed pair, y_i and y_j are two fixed values, so "their covariance" only has meaning as an average over an ensemble of pairs. The ensemble we use is the set of all pairs that share a given piece of *relatedness information* $\mathcal{I}_{ij}$ about the pair, and the object of interest is the conditional cross-product

$$C_{ij}(\mathcal{I}) \; := \; \mathrm{E}[\, y_i y_j \mid \mathcal{I}_{ij} \,]. \tag{2}$$

If $\mathrm{E}[y_i \mid \mathcal{I}] = 0$, then $C_{ij}(\mathcal{I}) = \mathrm{Cov}[y_i, y_j \mid \mathcal{I}]$. For most information $\mathcal{I}$, like the realized relatedness or degree between two relatives, this equivalence is applicable. An exception is when the information we are modelling contains individuals' actual genotype information, for which $\mathrm{E}[y_i \mid \text{genotypes}] = g_i \neq 0$, therefore the expected cross-product of two individuals is not equivalent to their covariance given the fixed information.

Using (1) together with **[No-IGE]** and **[Env-indep]**, every cross term involving e vanishes in expectation (each factors through a mean-zero environmental term), so the phenotypic cross-product reduces to the genetic one:

$$C_{ij}(\mathcal{I}) \;=\; \mathrm{E}[\,(g_i + e_i)(g_j + e_j) \mid \mathcal{I}\,] \;=\; \mathrm{E}[\,g_i g_j \mid \mathcal{I}\,]. \tag{3}$$

The rest of the article evaluates (3) at four levels of resolution for $\mathcal{I}$.

1.3 Level 0 — the exact genetic cross-product

At the finest resolution we condition on the realized genotypes of i and j at the causal variants, for which we know the true causal effect sizes. Then g_i and g_j are fixed, and by **[Additivity]**,

$$\boxed{C_{ij} \;=\; \mathrm{E}[\,y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j, \boldsymbol{\beta}\,] \;=\; g_i g_j \;=\; \sum_k \sum_{k'} \beta_k \beta_{k'}\, x_{ik}\, x_{jk'}} \tag{4}$$

Under our model and assumptions so far, this returns the true expected cross-product of two individuals' phenotypes. The genetic cross-product between two individuals is, fundamentally, an effect-size-weighted function of the genotypes they actually carry at causal sites. Notably, this formula does not require any knowledge of the relatedness of the two individuals.

1.4 Level 1 — genome-wide allelic sharing

If we do not know the causal effect sizes, then we cannot apply (4). However, we can assume that genetic effects' cross-products are independent of individuals' allelic sharing patterns.

[Random-effects] Random effects, exchangeable across variants

Treating the effects as random, they are mean-zero and uncorrelated across variants, the distribution of their cross-products $\{\beta_k\beta_{k'}\}$ is independent of the genotypes $\mathbf{X}$, and every variant carries the same second moment:

$$\mathrm{E}[\beta_k] = 0, \quad \mathrm{E}[\beta_k\beta_{k'}] = 0 \ \ (k \neq k'), \quad \mathrm{E}[\beta_k\beta_{k'} \mid \mathbf{X}] = \mathrm{E}[\beta_k\beta_{k'}], \quad \mathrm{E}[\beta_k^2] = \mathrm{E}[\beta_{k'}^2].$$

The last condition is a statement about the *prior* we place on effects, not about their realized values: it says we have no information distinguishing one variant from another, so we weight them symmetrically.

Commonly bundled into: the infinitesimal / random-effects model.

Taking the expectation of (4) over the effects, and using the independence of the cross-products from the genotypes (**[Random-effects]**) to replace $\mathrm{E}[\beta_k\beta_{k'} \mid \mathbf{X}]$ by $\mathrm{E}[\beta_k\beta_{k'}]$:

$$\mathrm{E}[\,g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j\,] = \sum_k \sum_{k'} \mathrm{E}[\beta_k\beta_{k'}]\, x_{ik}x_{jk'}.$$

Because the effects are mean-zero and uncorrelated across variants, $\mathrm{E}[\beta_k\beta_{k'}] = 0$ whenever $k \neq k'$; the entire off-diagonal ($k \neq k'$) collapses regardless of any correlation between the genotypes. What remains is the diagonal, weighted by each variant's effect variance:

$$\mathrm{E}[\,g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j\,] = \sum_k \mathrm{E}[\beta_k^2]\, x_{ik}x_{jk}.$$

The additive variance is the total of these: since $\mathrm{Var}[x_{ik}] = 1$ and the off-diagonal effect terms vanish, $V_A = \mathrm{Var}[g_i] = \sum_k \mathrm{E}[\beta_k^2]$, and thus $\mathrm{E}[\beta_k^2] = V_A/M$. Effectively, the per-variant effect weighting of (4) has been replaced by a uniform weighting across all variants. Therefore

$$C_{ij}(\mathbf{x}_i, \mathbf{x}_j) \;=\; V_A \cdot \frac{1}{M}\sum_k x_{ik}x_{jk}. \tag{5}$$

The quantity multiplying V_A is the genome–wide genotype cross-product (i.e. allelic sharing), which we'll define by:

$$A_{ij} = \frac{1}{M}\sum_k x_{ik}x_{jk}$$

This can be thought of as an identity-by-state (IBS)-based measure of "relatedness", and is exactly the (i, j) entry of what is typically computed as a Genetic Relatedness Matrix (GRM), $\mathbf{A}$. Therefore, we can further simplify:

$$\boxed{C_{ij}(A_{ij}) \;=\; V_A A_{ij}} \tag{6}$$

1.5 Level 2 — realized relatedness

The problem with calculating A_{ij} is that we may not have genotyped every causal variant that contributes to a trait. As long as some of the causal variants are not in perfect correlation with a genotyped variant, then the covariance of two individuals' phenotypes will not fully take into account the contributions of the ungenotyped causal variants. However, when individuals are related, their IBS-based relatedness from genotyped variants is informative of the IBS-based relatedness of ungenotyped variants. Therefore, the former can be used to capture the contributions of the latter.

When individuals inherit a genomic segment from a common ancestor, then all of the variants within that genomic segment will be identical-by-state, as long as we assume that de-novo mutations have not caused the inherited alleles to be different. These genomic segments and their variants are said to be identical-by-descent (IBD).

[No-mutation] IBD implies IBS

Alleles that are identical-by-descent are also identical-by-state: descent is transmitted without mutation.

Commonly bundled into: the infinite-sites / no-recurrent-mutation assumption.

When two individuals' alleles are IBD, then they are always IBS; when they are not IBD, they can still be IBS. Therefore, we can decompose the IBS-based relatedness A_{ij} into a sum over IBD allele pairs and a sum over non-IBD allele pairs. Write each standardized genotype as the sum of its two per-allele (maternal and paternal) standardized values, $x_{ik} = z_{ik}^{(m)} + z_{ik}^{(p)}$. Since $\mathrm{Var}[x_{ik}] = 1$ and the two alleles are independent ([No-inbreeding]), each per-allele value has $\mathrm{E}[z] = 0$ and $\mathrm{Var}[z] = \frac{1}{2}$. The genotype cross-product then expands over the four ordered allele pairs, and A_{ij} splits by IBD status:

$$\begin{aligned} A_{ij} &= \frac{1}{M}\sum_k x_{ik}x_{jk} = \frac{1}{M}\sum_k \sum_{u,v\in\{m,p\}} z_{ik}^{(u)} z_{jk}^{(v)} \\ &= \underbrace{\frac{1}{M}\sum_{\text{IBD pairs}} z_{ik}^{(u)} z_{jk}^{(v)}}_{\text{IBD}} + \underbrace{\frac{1}{M}\sum_{\text{non-IBD pairs}} z_{ik}^{(u)} z_{jk}^{(v)}}_{\text{non-IBD}}. \end{aligned} \tag{7}$$

For an IBD allele pair, the two per-allele values are copies of the same ancestral allele and, by [No-mutation], of the same state; they are therefore equal, so their expected cross-product is $\mathrm{E}[(z_{ik}^{(u)})^2] = \mathrm{Var}[z_{ik}^{(u)}] = \frac{1}{2}$. For non-IBD allele pairs, we have to make an assumption on how they are distributed.

[Non-IBD-indep] Independence of non-IBD alleles

Any two alleles that are *not* IBD are independent draws from the population allele distribution.

Commonly bundled into: between individuals: random mating, no population structure, and no assortative mating.

[No-inbreeding] **No inbreeding**

Within an individual, the two homologous alleles are not IBD with each other, and hence (by [Non-IBD-indep]) independent. This results in each variant's genotype variance being $2p_k(1-p_k)$ — so the standardized x_{ik} has variance 1 — and it keeps the number of alleles a pair shares IBD at a variant in $\{0,1,2\}$.

Commonly bundled into: a randomly mating, outbred population (Hardy–Weinberg within individuals).

By the definition of independent draws, the expected cross-product of a non-IBD allele pair is therefore $\mathrm{E}[z_{ik}^{(u)}]\,\mathrm{E}[z_{jk}^{(v)}] = 0$. Letting $\mathrm{IBD}_{ij}(k) \in \{0,1,2\}$ denote the number of alleles the pair shares IBD at variant k, exactly $\mathrm{IBD}_{ij}(k)$ of the four ordered allele pairs are IBD — each contributing $\frac{1}{2}$ — and the rest contribute 0, so

$$\mathrm{E}[\, x_{ik}x_{jk} \,|\, \mathrm{IBD}_{ij}(k)\,] = \tfrac{1}{2}\mathrm{IBD}_{ij}(k).$$

A variant thus contributes expected cross-product 0, $\frac{1}{2}$, or 1 according to whether the pair shares 0, 1, or 2 alleles IBD there. We define the pair's *realized relatedness* π_{ij} as the average IBD state across variants, halved — equivalently, the fraction of the pair's genome that is IBD:

$$\pi_{ij} \;:=\; \frac{1}{2M}\sum_k \mathrm{IBD}_{ij}(k). \tag{8}$$

Conditioning on the pair's realized IBD states and using the per-variant identity $\mathrm{E}[x_{ik}x_{jk} \mid \mathrm{IBD}_{ij}(k)] = \frac{1}{2}\,\mathrm{IBD}_{ij}(k)$, linearity of expectation gives the expected allelic sharing:

$$\mathrm{E}[\, A_{ij} \,|\, \{\mathrm{IBD}_{ij}(k)\}\,] = \frac{1}{M}\sum_k \tfrac{1}{2}\,\mathrm{IBD}_{ij}(k) = \frac{1}{2M}\sum_k \mathrm{IBD}_{ij}(k) = \pi_{ij}.$$

This depends on the IBD states only through π_{ij}, so we may condition on π_{ij} alone. Crucially, it required no assumption that the per-variant IBD states are independent — and indeed they are not: an IBD state changes only at a cross-over within a co-inherited segment, so physically linked variants have strongly correlated IBD states. Linearity of expectation is insensitive to this correlation.

Combining with (6), the pair's expected phenotypic cross-product is their realized relatedness scaled by the additive variance:

$$\boxed{\mathrm{E}[\, y_i y_j \mid \pi_{ij}\,] \;=\; V_A\,\mathrm{E}[A_{ij} \mid \pi_{ij}] \;=\; V_A\,\pi_{ij}}\,. \tag{9}$$

1.6 Level 3 — genetic degree

In cases where we cannot measure π_{ij} based on per-variant IBD states, we can still estimate it if we know the genetic degree d_{ij} between the individuals.

[Genetic-degree] Degree determines expected relatedness

The genetic degree d_{ij} fixes the distribution of IBD sharing through Mendelian segregation, so that the expected relatedness given the degree is a single number $\bar{\pi}_{d_{ij}}$:

$$\mathrm{E}[\pi_{ij} \mid d_{ij}] = \bar{\pi}_{d_{ij}}.$$

Furthermore, under the outbreeding assumption ([No-inbreeding]) — applied here not only to i and j but to the entire pedigree connecting them, so that it contains no inbreeding loops — $\bar{\pi}_{d_{ij}} = 2^{-d_{ij}}$, which is not fully derived here. We can then obtain the expected phenotypic covariance given the degree d_{ij}:

$$\mathrm{E}[\, y_i y_j \mid d_{ij} \,] \;=\; \mathrm{E}\big[\, \mathrm{E}[y_i y_j \mid \pi_{ij}] \,\big|\, d_{ij} \,\big] \;=\; V_A \, \mathrm{E}[\pi_{ij} \mid d_{ij}] \;=\; V_A \, \bar{\pi}_{d_{ij}}. \tag{10}$$

1.7 Summary

Each level conditions on coarser information than the one above and is its conditional expectation; the rightmost column lists the assumption(s) newly required to descend one step.

Explicitly, each level is the conditional expectation of the finer level above it, averaging out one additional layer of information:

$$\begin{aligned}
\mathrm{E}[y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j] &= \mathrm{E}\big[\, g_i g_j \,\big|\, \mathbf{x}_i, \mathbf{x}_j \,\big] = V_A \, A_{ij}, \\
\mathrm{E}[y_i y_j \mid \pi_{ij}] &= \mathrm{E}\big[\, V_A \, A_{ij} \,\big|\, \pi_{ij} \,\big] = V_A \, \pi_{ij}, \\
\mathrm{E}[y_i y_j \mid d_{ij}] &= \mathrm{E}\big[\, V_A \, \pi_{ij} \,\big|\, d_{ij} \,\big] = V_A \, \bar{\pi}_{d_{ij}},
\end{aligned}$$

starting from the Level-0 value $\mathrm{E}[y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j, \boldsymbol{\beta}] = g_i g_j$. Equivalently, the ladder telescopes into a single nested expectation,

$$\mathrm{E}[y_i y_j \mid d_{ij}] = \mathrm{E}\Big[\, \mathrm{E}\big[\, \mathrm{E}[\, g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j \,] \,\big|\, \pi_{ij} \,\big] \,\Big|\, d_{ij} \,\Big] = V_A \, \bar{\pi}_{d_{ij}}.$$

This is valid down the generative chain $d_{ij} \to \pi_{ij} \to (\mathbf{x}_i, \mathbf{x}_j) \to y$: conditioned on the finer variable, the phenotype does not depend further on the coarser one, so the tower property applies at each step.

Level	**Conditioned on**	$\mathrm{E}[y_i y_j \mid \cdot]$	**New assumptions**
0	causal genotypes	$g_i g_j$	[Additivity] (given [No-IGE], [Env-indep])
1	realized sharing A_{ij}	$V_A \, A_{ij}$	[Random-effects]
2	realized relatedness π_{ij}	$V_A \, \pi_{ij}$	[No-mutation], [Non-IBD-indep], [No-inbreeding]
3	genetic degree d_{ij}	$V_A \, \bar{\pi}_{d_{ij}}$	[Genetic-degree]

The property that makes this a ladder is worth naming, because the next architecture breaks it. At each step, the finer variable *screens off* the coarser one: once we condition on π_{ij}, knowing d_{ij} as well would tell us nothing more about $y_i y_j$, and likewise for the other steps. Screening is what licenses the tower property, and it is the reason the four information sets can be arranged in a single descending chain rather than a more complicated structure.

2 Additive genetics under assortative mating

We will now do away with the assumption of random mating (particularly implied in **[Non-IBD-indep]**) and instead provide a model for assortative mating that will let us quantify its impact on phenotypic covariance.

The phenotype model remains the same: we assume an additive genetic architecture (**[Additivity]**) with an independent environmental component (**[GE-indep]**). As a consequence of the assortative mating model we adopt below, covariance between the environmental and genetic components between some individuals will arise, breaking the weakest forms of the assumptions **[No-IGE]** and **[Env-indep]**. We also still assume no new mutations (**[No-mutation]**) and no inbreeding (**[No-inbreeding]**).

2.1 The mating model

We assume a model of assortative mating based on mates' similarity in a sorting factor, which is also our focal trait.

[Primary-AM] Primary Assortative Mating

Mating is non-random such that two mates' phenotypes are correlated for some trait at the time of mating:

$$\mathrm{Corr}[\,y_m, y_p\,] = \rho_y,$$

where the subscripts m and p index the maternal and paternal member of a mated pair. Matching is on the phenotype *only*: any covariance between the mates' trait components is a downstream consequence of this phenotypic matching, and is not specified separately. Following the convention that a subscript names the component being correlated, we write ρ_g for the induced correlation between mates' genetic values: $\mathrm{Corr}[\,g_m, g_p\,] = \rho_g^{(t)}$, where the superscript records that this induced correlation changes from one generation to the next, unlike ρ_y, which is held fixed across generations.

Commonly bundled into: primary phenotypic assortment, as opposed to social homogamy (matching on a shared environment) or genetic homogamy (matching on genotype directly); it also excludes convergence, where mates grow more similar after mating. ρ_y is often written μ or r in the literature.

2.2 Dynamics and equilibrium

When starting from a randomly mating population, one generation of assortative mating causes the immediate offspring generation to exhibit different phenotypic and genetic properties on expectation. Luckily, if the parameters of assortative mating remain fixed, an equilibrium is reached such that stable properties of the population can be studied. We will first explore how assortative mating affects statistical properties of the population during the first few generations, then we will solve for the parameter values that lead to an equilibrium state.

2.2.1 Path Analysis

First, we'll introduce path analysis, a very helpful tool for modelling the covariance between variables in a causal system, such as genetic inheritance and assortative mating. Path analysis was introduced by Sewall Wright [1], as a way of decomposing an observed correlation between variables into the contributions of a hypothesized causal structure, and set out systematically thirteen years later as the method of path coefficients [2]. Wright applied it to mating systems in the same period [3, 4]; together with Fisher's treatment of the same problem [5], that work is the origin of the classical results on how assortment changes the correlation between relatives — the results this section rederives. The explanations below are adapted from the Supplementary Note of Sunde et al. [6], which introduced the co-path edge type (described below). Unlike Sunde et al. [6], we do not assume that the variables are standardized, so we will keep track of variances explicitly.

A *path diagram* is a graphical representation of a generative model. Each variable is a node — drawn as a rectangle if it is observed and an ellipse if it is latent — and the nodes are joined by three kinds of edges:

- A **directed** edge $a \to b$ labelled with a coefficient c says that a is a cause of b, contributing $c\,a$ to b's structural equation. Collecting the arrows that point *into* a node recovers that node's equation, so the diagram and the model equations carry the same content.
- A **bidirected** edge $a \leftrightarrow b$ carries covariance between the *disturbances* of a and b. A variable's disturbance is what remains of it once its causes in the diagram are accounted for: if the arrows into b give $b = c_1 a_1 + \cdots + c_k a_k + \epsilon_b$, then ϵ_b is b's disturbance, and for a variable with no incoming arrows the disturbance is the variable itself. The self-loop $a \leftrightarrow a$ is the case $b = a$: it is $\mathrm{Var}[\epsilon_a]$, the part of $\mathrm{Var}[a]$ not supplied by a's causes, so a variable with no incoming arrows carries its entire variance on such a loop.
- A **co-path** a — b, drawn as a plain line with no arrowheads, is covariance attributable to *matching*: a and b came to be associated because they were paired on their values, not because they share a cause.

The covariance between *any* two variables in the diagram, latent or observed, can be calculated quickly by following Wright's tracing rules. Enumerate every *valid chain* joining the two variables; their covariance is the sum over chains of the product of the coefficients each chain passes through. A valid chain between a and b has the shape

$$a \leftarrow \cdots \leftarrow u \ \leftrightarrow \ v \rightarrow \cdots \rightarrow b:$$

trace *backward* along directed edges from a to some ancestor u, cross *exactly one* bidirected edge $u \leftrightarrow v$ (possibly a self-loop, $u = v$), then trace *forward* along directed edges from v down to b. A chain turns around exactly once, and it turns around on a bidirected edge, because that is the only place variance enters the system.

Co-paths extend this. A chain may pass *through* a co-path and keep going: a co-path joins two valid chains of the kind above into a longer one,

$$\underbrace{a \leftarrow \cdots \leftrightarrow \cdots \rightarrow y_m}_{\text{valid chain}} - \underbrace{y_p \leftarrow \cdots \leftrightarrow \cdots \rightarrow b}_{\text{valid chain}}.$$

Two further rules apply [6]: a chain can neither start nor end with a co-path, and a chain cannot use two co-paths belonging to the same mating process. It *may* cross co-paths of different mating processes.

A bidirected edge is a statement about disturbances, so a chain terminates on it: a disturbance has no causes in the diagram, so there is nothing to keep tracing back through, and the chain cannot reach past y_m to the things that caused y_m. A co-path is a statement about the matched variables themselves, so a chain continues backward through it, which encodes the fact that matching two people on their phenotypes induces correlation between *all the causes* of those phenotypes — their genetic values, and every individual variant behind them. This is the mathematical basis behind the **[Primary-AM]** assumption.

The quantity multiplied along a chain when it crosses a co-path is not the correlation ρ_y itself, but

$$\mu^{(t)} = \frac{\rho_y}{\sqrt{\mathrm{Var}[y_m^{(t)}]\,\mathrm{Var}[y_p^{(t)}]}} = \frac{\rho_y}{V_P^{(t)}}, \tag{11}$$

because a co-path with coefficient μ contributes $\mathrm{Cov}[a, b] = \mu\,\mathrm{Var}[a]\,\mathrm{Var}[b]$ [6]. The two coincide only when the matched variables have unit variance. Note the use of a superscript to denote the generation a quantity belongs to, since $V_P^{(t)}$ changes from one generation to the next under assortative mating (described below). A value of μ carried unchanged from one generation to the next is therefore wrong, which is the practical reason we label a co-path with its correlation in brackets, as $[\rho_y]$, and leave $\mu^{(t)}$ to be derived from the diagram's own variances.

Tracing chains by hand is instructive for a single pair and hopeless for a pedigree, since the number of chains grows multiplicatively with depth. The diagrams and covariances in this section were produced with `pathMgr`, a small package written alongside this writeup: it takes a model specification, returns the covariance between any two of its variables as an algebraic expression, and emits the path diagram to go with it. It computes each covariance in two independent ways — by enumerating chains exactly as above, and in closed form from the matrix identity $\mathbf{\Sigma} = \mathbf{F}(\mathbf{I}-\mathbf{A})^{-1}\mathbf{S}(\mathbf{I}-\mathbf{A})^{-\intercal}\mathbf{F}^{\intercal}$ — and their agreement is the correctness check.

That identity is the diagram written as matrices, one row and column per variable. Collecting all the variables into a vector $\mathbf{v}$, the matrix $\mathbf{A}$ holds the **directed** edges — A_{ij}

is the coefficient on the arrow from j to i — so the whole system of structural equations is $\mathbf{v} = \mathbf{A}\mathbf{v} + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon}$ is the vector of disturbances. The symmetric matrix $\mathbf{S}$ holds the **bidirected** edges: S_{ij} is the value of $i \leftrightarrow j$, and its diagonal S_{ii} is the self-loop, the variance of i's disturbance. $\mathbf{F}$ merely selects rows, keeping the variables we want reported and discarding the rest.

The inverse is what does the tracing. In an acyclic model $\mathbf{A}$ is nilpotent, so $(\mathbf{I} - \mathbf{A})^{-1} = \mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \cdots$, and the (i, u) entry of that sum is the total over all directed paths from u to i of the product of their coefficients — one term per path, $\mathbf{A}^n$ contributing the paths of length n. Sandwiching $\mathbf{S}$ between it and its transpose therefore gives, for latent and observed variables alike,

$$\Sigma_{ij} = \sum_u \sum_v \underbrace{\left[(\mathbf{I} - \mathbf{A})^{-1}\right]_{iu}}_{i \leftarrow \cdots \leftarrow u} \underbrace{S_{uv}}_{u \leftrightarrow v} \underbrace{\left[(\mathbf{I} - \mathbf{A})^{-1}\right]_{jv}}_{v \rightarrow \cdots \rightarrow j},$$

which is Wright's rule term for term: a backward leg, exactly one bidirected edge, a forward leg, summed over every chain, indexed by the $u \leftrightarrow v$ it turns around on. The two methods are the same statement — one enumerated, one summed in closed form — which is why their agreement tests the implementation rather than the mathematics. Co-paths are the one thing that sits outside this identity and contributes further terms; there, the chain rules given above are the definition.

2.2.2 Consequences of assortative mating on first generation of mates

We assume a population has been randomly mating as described by assumption **[Non-IBD-indep]** up to some generation $t = 0$, which can also be referred to as the base or randomly-mating population. The parameters defined in the previous section, such as the additive variance V_A and heritability h^2, are measured in this base population and denoted by a (0), e.g. $V_A^{(0)}$ and $h^2_{(0)}$, since these terms do not remain fixed across generations under assortative mating. Note that V_E remains fixed.

Building the model from haploid alleles

To properly model the effects of assortative mating, it pays off to think in terms of individual haploid alleles, rather than diploid genotypes. We previously defined $z_{ik}^{(m)}$ and $z_{ik}^{(p)}$ as the standardized values of the maternally- and paternally-inherited alleles at variant k for individual i, and $x_{ik} = z_{ik}^{(m)} + z_{ik}^{(p)}$ as the standardized diploid genotype. Each haploid allele $a^{(m)}$ and $a^{(p)}$ can take on two values, 0 or 1, and therefore each has a variance of $p_k(1 - p_k)$, where p_k is the allele frequency at variant k. Typically, diploid genotypes are standardized by their expected variance when the two alleles are independent, which is $2p_k(1 - p_k)$, resulting in a standardized variance of 1, as used in the previous section. Under this standardization, each haploid allele has a standardized variance of $\frac{1}{2}$, and if we assume that the allele frequency does not change over time, then neither does this quanity:

[Constant-freq] Allele frequencies do not change

The allele frequency p_k of each variant is the same across every generation. As a consequence, the standardization divides by a fixed quantity, and the standardized variance of each haploid allele is the same in every generation:

$$z_{ik} = \frac{a_{ik} - p_k}{\sqrt{2p_k(1-p_k)}} \quad \Longrightarrow \quad \mathrm{Var}[z_{ik}] = \frac{\mathrm{Var}[a_{ik}]}{2p_k(1-p_k)} = \frac{p_k(1-p_k)}{2p_k(1-p_k)} = \tfrac{1}{2},$$

where $a_{ik} \in \{0, 1\}$ is the unstandardized haploid allele count and p_k is the allele frequency at variant k.

Commonly bundled into: no selection on the trait, no drift, no migration; in a finite population it is the large-N_e limit [7].

In contrast, the standardized diploid genotype can have non-unit variance and change over time if the two alleles are correlated, as will be demonstrated in the case of assortative mating. Our path diagram will therefore start with haploid alleles which are summed together to form diploid genotypes, which are then summed to form the additive genetic value, which is then added to the environmental value to form the phenotype.

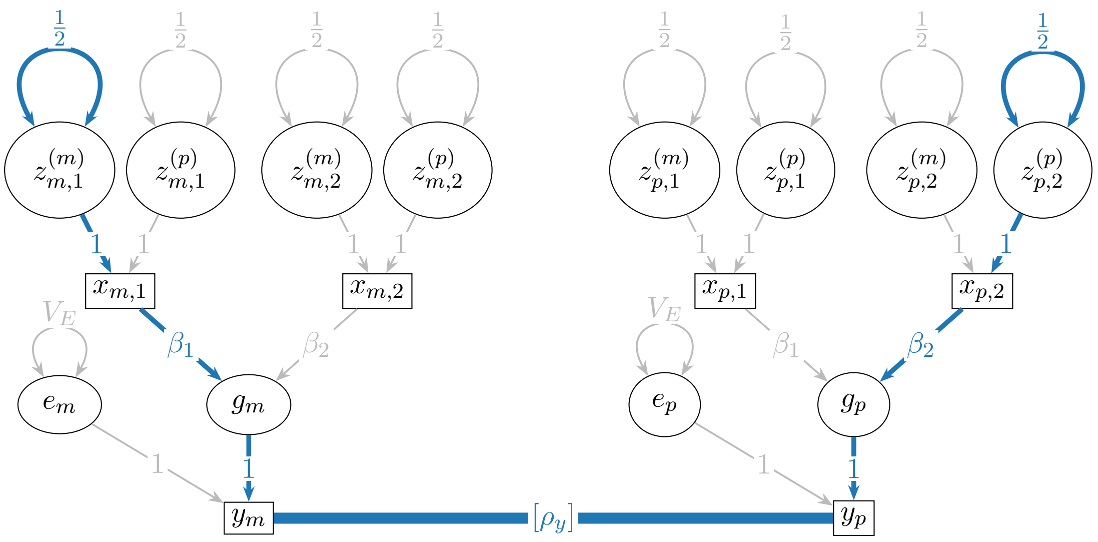
The diagram

Mates in generation $t = 0$ are formed according to **[Primary-AM]**. Below is the path diagram for a single pair that assortatively mated on the focal trait with strength ρ_y. The trait's genetic component is the additive sum of two causal variants, which is the smallest example that shows a *cross-variant* covariance; the formulas we state work for any number of variants.

One chain is traced on the diagram to show the rules in use. It runs from a single allele in one mate to a single allele at a *different* variant in the other — a covariance that is zero under random mating, and for which there is no common ancestor to appeal to. Multiplying the coefficients along it gives

$$\mathrm{Cov}[z^{(m)}_{m,1},\, z^{(p)}_{p,2}] = \tfrac{1}{2} \cdot \beta_1 \cdot \mu^{(0)} \cdot \beta_2 \cdot \tfrac{1}{2} = \frac{\beta_1 \beta_2\, \rho_y}{4\, V_P^{(0)}},$$

one $\frac{1}{2}$ for each allele's share of its genotype, and one effect size for each of the *two* variants involved — not the square of either. There are four such allele pairs behind any one pair of genotypes, and they are identical, so they sum back to $\mathrm{Cov}[x_{m,1}, x_{p,2}] = \beta_1\beta_2\rho_y/V_P^{(0)}$.



$$z^{(m)}_{m,1} \leftrightarrow z^{(m)}_{m,1} \to x_{m,1} \to g_m \to y_m \text{ — } y_p \leftarrow g_p \leftarrow x_{p,2} \leftarrow z^{(p)}_{p,2} \leftrightarrow z^{(p)}_{p,2}$$

$$\mathrm{Cov}\left[z^{(m)}_{m,1}, z^{(p)}_{p,2}\right] = \tfrac{1}{2}\cdot 1\cdot \beta_1 \cdot 1 \cdot \frac{\rho_y}{V_P^{(0)}} \cdot 1 \cdot \beta_2 \cdot 1 \cdot \tfrac{1}{2} = \frac{\beta_1\beta_2\rho_y}{4V_P^{(0)}}$$

Figure 1: A mated pair in the base population, assorting on the phenotype with strength ρ_y, with one chain traced on it and the product of its coefficients written underneath. The model is built from the haploid alleles: $z^{(m)}_{i,k}$ and $z^{(p)}_{i,k}$ are the maternally and paternally inherited alleles of individual i at variant k, each with variance $\frac{1}{2}$, and the genotype $x_{i,k} = z^{(m)}_{i,k} + z^{(p)}_{i,k}$. Then $g_i = \beta_1 x_{i,1} + \beta_2 x_{i,2}$ and $y_i = g_i + e_i$. Genotypes and phenotypes are observed (rectangles); alleles, genetic and environmental values are latent (ellipses) — an allele's parental origin is not read off an unphased genotype. The single thick line is the co-path, labelled with the correlation it induces. The edges the chain does not use are faded rather than omitted, so the figure still states the whole model and any other covariance can be traced off the same picture.

Doing the same for every pair of variables in the diagram gives the two tables below. They are stated at the genotype level, since that is the scale the rest of the section works at; the allele level adds only the four facts already used above, all of which hold in the base population and none of which survive into the next generation:

$$\mathrm{Var}\big[z^{(u)}_{i,k}\big] = \tfrac{1}{2}, \qquad \mathrm{Cov}\big[z^{(u)}_{i,k},\, z^{(v)}_{i,l}\big] = 0 \;\text{ for } (u,k) \neq (v,l), \qquad \mathrm{Cov}\big[z^{(u)}_{i,k},\, y_i\big] = \tfrac{\beta_k}{2},$$

for $u, v \in \{m, p\}$ naming the parental origin. The middle one is two statements at once: an individual's own two alleles at the same variant are uncorrelated ($k = l$, the quantity named α below), and there is no gametic phase disequilibrium between different variants ($k \neq l$).

Within an individual (Table 1) nothing has changed from the randomly mating case: these are base-population individuals, so their genotypes are still uncorrelated with each other and with their own environment.

	$x_{i,1}$	$x_{i,2}$	g_i	e_i	y_i
$x_{i,1}$	1	0	β_1	0	β_1
$x_{i,2}$	0	1	β_2	0	β_2
g_i	β_1	β_2	$V_A^{(0)}$	0	$V_A^{(0)}$
e_i	0	0	0	V_E	V_E
y_i	β_1	β_2	$V_A^{(0)}$	V_E	$V_P^{(0)}$

Table 1: Covariances within one individual $i \in \{m, p\}$, identical for both members of the pair. Here $V_A^{(0)} = \beta_1^2 + \beta_2^2$ and $V_P^{(0)} = V_A^{(0)} + V_E$. The last column — each variable's covariance with its own phenotype — is the only part of this table needed for computing the covariance between variables across mates.

The co-path induces covariance between the factors pertaining to mates such that:

$$\boxed{\mathrm{Cov}[\, a_m,\, b_p\,] \;=\; \mathrm{Cov}[\, a_m,\, y_m\,] \cdot \mu^{(0)} \cdot \mathrm{Cov}[\, y_p,\, b_p\,]} \tag{12}$$

for any variables a and b belonging to the maternal and paternal member respectively. The entire cross-mate block is therefore the outer product of a single vector — the last column of Table 1 — scaled by $\mu^{(0)} = \rho_y / V_P^{(0)}$.

$\times\, \mu^{(0)}$	$x_{p,1}$	$x_{p,2}$	g_p	e_p	y_p
$x_{m,1}$	β_1^2	$\beta_1\beta_2$	$\beta_1 V_A^{(0)}$	$\beta_1 V_E$	$\beta_1 V_P^{(0)}$
$x_{m,2}$	$\beta_1\beta_2$	β_2^2	$\beta_2 V_A^{(0)}$	$\beta_2 V_E$	$\beta_2 V_P^{(0)}$
g_m	$\beta_1 V_A^{(0)}$	$\beta_2 V_A^{(0)}$	$(V_A^{(0)})^2$	$V_A^{(0)} V_E$	$V_A^{(0)} V_P^{(0)}$
e_m	$\beta_1 V_E$	$\beta_2 V_E$	$V_A^{(0)} V_E$	V_E^2	$V_E V_P^{(0)}$
y_m	$\beta_1 V_P^{(0)}$	$\beta_2 V_P^{(0)}$	$V_A^{(0)} V_P^{(0)}$	$V_E V_P^{(0)}$	$(V_P^{(0)})^2$

Table 2: Covariances between the two mates. **Every entry carries a factor** $\mu^{(0)} = \rho_y / V_P^{(0)}$, shown once in the corner rather than repeated in twenty-five cells. The block is symmetric, and is the outer product of $(\beta_1, \beta_2, V_A^{(0)}, V_E, V_P^{(0)})$ with itself — a direct consequence of (12). Setting $\rho_y = 0$ returns every entry to zero, recovering the randomly mating case of Section 1.

Three key results from Table 2 are highlighted here. We also generalize from two variants to M variants, where $g_i = \sum_{k=1}^{M} \beta_k x_{ik}$.

- $\mathrm{Cov}[\, x_{mk},\, x_{pl}\,] \;=\; \beta_k \beta_l\, \rho_y / V_P^{(0)} \;\neq\; 0$, for *every* pair of variants k and l. Alleles at different loci in different individuals are correlated, signed by the product of their effects and breaking the **[Non-IBD-indep]** assumption Note that in the base population, the within-individual covariance $\mathrm{Cov}[\, x_{mk},\, x_{ml}\,]$ is still exactly zero for $k \neq l$; the disequilibrium exists only *between* the mates.
- $\mathrm{Cov}[\, e_m,\, g_p\,] = V_A^{(0)} V_E\, \rho_y / V_P^{(0)} \neq 0$. One mate's environment is correlated with the other's genetic value, which is the break of **[Env-indep]** and **[No-IGE]** flagged at the start of this section. Despite one mate's genotype not causally affecting the other's environment, the two still become correlated due to the phenotypic matching.

– $\rho_g^{(0)} = \mathrm{Corr}[\, g_m,\, g_p\,] = \frac{(V_A^{(0)})^2 \rho_y / V_P^{(0)}}{V_A^{(0)}} = \rho_y \frac{V_A^{(0)}}{V_P^{(0)}} = \rho_y\, h^2_{(0)}$. Mates' genetic values are correlated according to the strength of assortative mating and the heritability of the sorting trait. Equivalently, this is the sum of the first result over all M^2 ordered pairs of variants, weighted by their effects: $\sum_k \sum_l \beta_k\beta_l \cdot \beta_k\beta_l\rho_y/V_P^{(0)} = (\sum_k \beta_k^2)^2\rho_y/V_P^{(0)}$.

2.2.3 Consequences of assortative mating on first generation of offspring

Next, we will extend our path diagram to contain two offspring of an assortatively mated pair. For each variant, all offspring inherit one allele from each parent, but which allele they inherit from that parent (the maternal vs paternal allele) is uniformly random. For the purposes of tracking covariances, we model this by drawing an edge from each parental allele to their respective inherited allele in the offspring and adding an exogeneous source of variance. The directed edges have a coefficient of $\frac{1}{2}$ to reflect that the expected inherited allele in the offspring is the average of the two alleles of the parent being inherited from.

Write o for an offspring of the mated pair, and let $B_{ok}^{(m)} \in \{0, 1\}$ indicate which of the mother's two alleles was transmitted at variant k. We arbitrarily label the maternal allele $z_{mk}^{(m)}$ and the paternal allele $z_{mk}^{(p)}$ such that if $B_{ok}^{(m)} = 1$, then the offspring inherits the maternal allele, and if $B_{ok}^{(m)} = 0$ then the offspring inherits the paternal allele.

$$z_{ok}^{(m)} \;=\; B_{ok}^{(m)}\, z_{mk}^{(m)} \;+\; \big(1 - B_{ok}^{(m)}\big)\, z_{mk}^{(p)}, \qquad B_{ok}^{(m)} \sim \mathrm{Bernoulli}(\tfrac{1}{2}), \quad B_{ok}^{(m)} \perp z_{mk}. \tag{13}$$

A path diagram represents the *linear projection* of the random variable: the best prediction of the offspring's allele from the mother's two, which is their average,

$$\mathrm{E}\big[\, z_{ok}^{(m)} \,\big|\, z_{mk}^{(m)}, z_{mk}^{(p)}\,\big] \;=\; \tfrac{1}{2} z_{mk}^{(m)} + \tfrac{1}{2} z_{mk}^{(p)} \;=\; \tfrac{1}{2} x_{mk}. \tag{14}$$

The exogenous variance is what the projection leaves behind. Subtracting (14) from (13) gives the *segregation deviation*

$$s_{ok}^{(m)} \;:=\; z_{ok}^{(m)} - \tfrac{1}{2} x_{mk} \;=\; \big(B_{ok}^{(m)} - \tfrac{1}{2}\big)\big(z_{mk}^{(m)} - z_{mk}^{(p)}\big), \tag{15}$$

which is mean-zero and uncorrelated with both of the mother's alleles. Squaring (15) and using $\mathrm{Var}\big[B_{ok}^{(m)}\big] = \frac{1}{4}$ gives its variance:

$$\mathrm{Var}\big[s_{ok}^{(m)}\big] \;=\; \tfrac{1}{4}\,\mathrm{E}\big[(z_{mk}^{(m)} - z_{mk}^{(p)})^2\big] \;=\; \tfrac{1}{4}\Big(\tfrac{1}{2} + \tfrac{1}{2} - 2\,\mathrm{Cov}\big[z_{mk}^{(m)}, z_{mk}^{(p)}\big]\Big) \;=\; \tfrac{1}{4} - \tfrac{1}{2}\,\mathrm{Cov}\big[z_{mk}^{(m)}, z_{mk}^{(p)}\big]. \tag{16}$$

The offspring's paternally inherited allele $z_{ok}^{(p)}$ is built the same way from the father's two alleles, with its own coin $B_{ok}^{(p)}$ and its own residual $s_{ok}^{(p)}$. Every coin is independent of every other, across parental origins and across offspring: a second child of the same pair draws its own $B_{o'k}^{(m)}$ and $B_{o'k}^{(p)}$, which is what makes two siblings differ. Following Yengo et al. [7] we define $\alpha_k^{(t)}$ as the *correlation* of the maternally and paternally inherited alleles of an individual i of generation t at variant k:

$$\alpha_k^{(t)} \;=\; \mathrm{Corr}\big[z_{ik}^{(m)},\, z_{ik}^{(p)}\big] \quad\Longrightarrow\quad \mathrm{Cov}\big[z_{ik}^{(m)},\, z_{ik}^{(p)}\big] \;=\; \tfrac{1}{2}\alpha_k^{(t)}. \tag{17}$$

Of particular importance is the variance of the diploid genotype, which is inflated by $\alpha_k^{(t)}$:

$$\begin{aligned}\mathrm{Var}[x_{ik}] &= \mathrm{Var}[z_{ik}^{(m)} + z_{ik}^{(p)}] \\ &= \mathrm{Var}[z_{ik}^{(m)}] + \mathrm{Var}[z_{ik}^{(p)}] + 2\,\mathrm{Cov}[z_{ik}^{(m)}, z_{ik}^{(p)}] \\ &= \tfrac{1}{2} + \tfrac{1}{2} + 2 \cdot \tfrac{\alpha_k^{(t)}}{2}\end{aligned} \tag{18}$$

$$\boxed{\mathrm{Var}[x_{ik}] \;=\; 1 + \alpha_k^{(t)}}$$

In the base population there is no inbreeding and mating is random, so $\alpha_k^{(0)} = 0$ for every variant, and everything reduces to the familiar $\mathrm{Var}[x_{ik}] = 1$.

Substituting (17) back into (16) puts the segregation variance in the same terms:

$$\boxed{\mathrm{Var}[s_{ok}^{(m)}] \;=\; \frac{1 - \alpha_k^{(t)}}{4}} \tag{19}$$

with $\alpha_k^{(t)}$ that of the transmitting parent. A parent whose two alleles are more alike contributes *less* segregation variance, because it matters less which of the two the offspring received; and in the base population, where $\alpha_k^{(0)} = 0$, it takes the familiar value $\frac{1}{4}$.

Below is a path diagram of two offspring of a mated pair that assortatively mated on the focal trait with strength ρ_y, for a trait with two causal variants. It is the diagram of Figure 1 with transmission added and nothing else; every allele covariance in the offspring generation is traced out of it rather than written into it.

We can use our path tracing rules to compute the correlation between maternally- and paternally-inherited alleles in the first offspring generation, $\alpha_k^{(1)}$:

Exactly four chains connect the offspring's two alleles at variant k. Each traces back from $z_{ok}^{(m)}$ to *one* of the mother's two alleles, turns around on that allele's variance, runs up through her genotype and phenotype, crosses the co-path, and comes back down the father's side to *one* of his two alleles and into $z_{ok}^{(p)}$. Two choices on each side give $2 \times 2 = 4$ chains, one for each $u, v \in \{m, p\}$:

$$\begin{aligned}&z_{ok}^{(m)} \leftarrow z_{mk}^{(u)} \leftrightarrow z_{mk}^{(u)} \rightarrow x_{mk} \rightarrow g_m \rightarrow y_m \\ &— \; y_p \leftarrow g_p \leftarrow x_{pk} \leftarrow z_{pk}^{(v)} \leftrightarrow z_{pk}^{(v)} \rightarrow z_{ok}^{(p)},\end{aligned}$$

broken across two lines at the co-path, which is exactly where it joins two valid chains into one. They are identical in value, since every allele carries the same variance $\frac{1}{2}$ and the same transmission coefficient $\frac{1}{2}$:

$$\underbrace{\tfrac{1}{2}}_{\text{transmission}} \cdot \underbrace{\tfrac{1}{2}}_{\mathrm{Var}[z]} \cdot \beta_k \cdot \mu^{(0)} \cdot \beta_k \cdot \underbrace{\tfrac{1}{2}}_{\mathrm{Var}[z]} \cdot \underbrace{\tfrac{1}{2}}_{\text{transmission}} = \frac{\beta_k^2\, \mu^{(0)}}{16}.$$

Summing the four and dividing by $\sqrt{\frac{1}{2} \cdot \frac{1}{2}} = \frac{1}{2}$ to convert the covariance to a correlation,

$$\alpha_k^{(1)} \;=\; \frac{4 \cdot \beta_k^2 \mu^{(0)}/16}{1/2} \;=\; \frac{\beta_k^2\, \mu^{(0)}}{2} \;=\; \boxed{\frac{\beta_k^2\, \rho_y}{2\, V_P^{(0)}}} \tag{20}$$

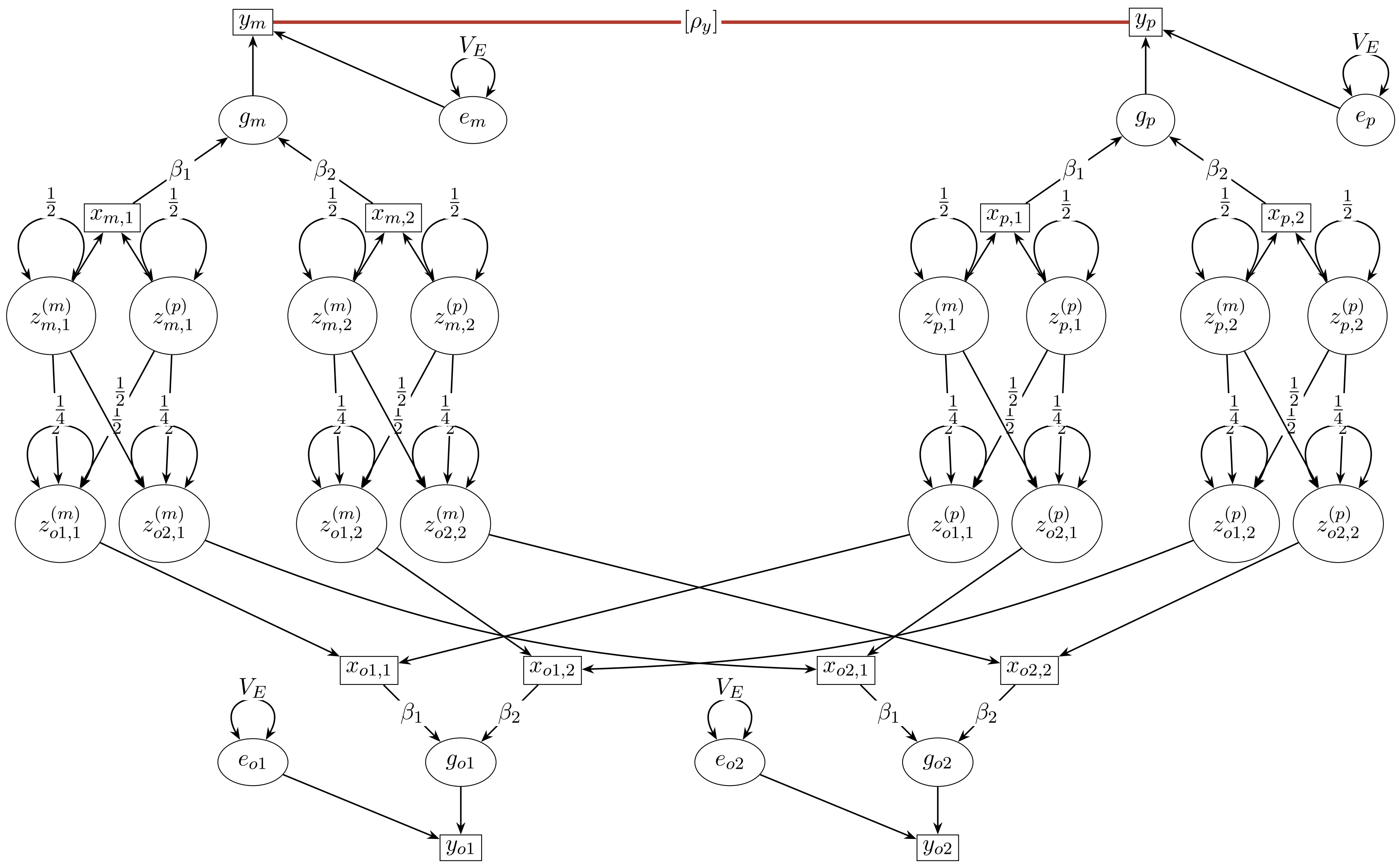


Figure 2: One mated pair from the base population and two of their offspring, at the allele level, for a trait with two causal variants. The parents are drawn *flowing upward* — alleles at the bottom of each block, phenotype at the top — so that the two generations' allele rows sit adjacent and the sixteen transmission edges stay short; the cost is that each offspring's $z \to x$ edges span the figure. Each inherited allele is fed by *both* alleles of the parent it came from, at $\frac{1}{2}$ each, per (14). The self-loop of $\frac{1}{4}$ on each offspring allele is its segregation variance, (19) evaluated at $\alpha_k^{(0)} = 0$: it is drawn as a disturbance variance because that is exactly what it is — the part of the allele's variance its causes in the diagram do not supply. The two offspring differ only through their coins being independent; nothing else distinguishes them, and e_{o1} and e_{o2} are separate draws. Note that no x anywhere carries a self-loop: $\mathrm{Var}[x] = 1 + \alpha_k^{(t)}$ is derived at both generations, and it takes the value 1 in the parents and a larger one in the offspring without a single edge differing between them.

We use the path tracing rules to compute the covariance between an individual and each of its two kinds of first-degree relative, at all three levels of the model. The results are collected in Table 3; they are stated for M variants, having been derived on the two-variant diagram. Every entry there turns out to be its randomly mating value plus a single quantity, so we name that quantity rather than write it out five times:

$$\Delta^{(t)} := \frac{\rho_y \left(V_A^{(t-1)}\right)^2}{2\, V_P^{(t-1)}} = \frac{V_A^{(t-1)}}{2}\, \rho_g^{(t-1)}, \tag{21}$$

the covariance that assortment adds at the genetic-value level among individuals of generation t. It carries a generation index for the same reason $\alpha_k^{(t)}$ does, and follows the same convention: both are named for the generation they *describe* while being computed from the variances of the parents who produced it, one generation back. For the first offspring generation, $\Delta^{(1)} = \rho_y (V_A^{(0)})^2/(2V_P^{(0)})$. Written this way nothing is hidden inside ρ_g: the increment is built only from ρ_y and the variances $V_A^{(0)}$ and $V_P^{(0)}$.

	self	full siblings	parent–offspring
genotypes, same variant k	$1 + \alpha_k^{(1)}$	$\frac{1}{2} + \alpha_k^{(1)}$	$\frac{1}{2} + \alpha_k^{(1)}$
genotypes, variants $k \neq l$	$\frac{\beta_k \beta_l\, \rho_y}{2\, V_P^{(0)}}$	$\frac{\beta_k \beta_l\, \rho_y}{2\, V_P^{(0)}}$	$\frac{\beta_k \beta_l\, \rho_y}{2\, V_P^{(0)}}$
genetic values	$V_A^{(0)} + \Delta^{(1)}$	$\frac{V_A^{(0)}}{2} + \Delta^{(1)}$	$\frac{V_A^{(0)}}{2} + \Delta^{(1)}$
phenotypes	$V_A^{(0)} + \Delta^{(1)} + V_E$	$\frac{V_A^{(0)}}{2} + \Delta^{(1)}$	$\frac{V_A^{(0)}}{2} + \frac{\rho_y V_A^{(0)}}{2}$

Table 3: Variances and covariances for an individual of the first generation produced under assortative mating, against itself and against each kind of first-degree relative: the "self" column is the pair (o_1, o_1), "full siblings" is (o_1, o_2), and "parent–offspring" is (m, o_1). Every entry is written as its randomly mating value plus what assortment added, and each added term is built only from ρ_y and the base-population variances: $\Delta^{(1)} = \rho_y (V_A^{(0)})^2/(2V_P^{(0)})$ from (21), and $\alpha_k^{(1)} = \beta_k^2 \rho_y/(2V_P^{(0)})$ from (20). Setting $\rho_y = 0$ kills both and returns the table to 1 or $\frac{1}{2}$, 0, and $V_A^{(0)}$ or $V_A^{(0)}/2$. **The added term is the same in every cell but one**: the bottom right, where it is $\rho_y V_A^{(0)}/2 = \Delta^{(1)}/h^2_{(0)}$, larger by a factor $1/h^2_{(0)}$.

Note that covariances for parents vs full-siblings only differ at the phenotypic level, and that the increment $\Delta^{(1)}$ is the same for both. The entire difference is one extra term available to the lineal pair and not to the collateral one:

$$\mathrm{Cov}[\, y_m, y_{o_1}\,] = \underbrace{\mathrm{Cov}[\, g_m, g_{o_1}\,]}_{\frac{V_A^{(0)}}{2}(1+\rho_g^{(0)})} + \underbrace{\mathrm{Cov}[\, e_m, g_{o_1}\,]}_{\frac{V_A^{(0)} V_E\, \rho_y}{2\, V_P^{(0)}}} = \frac{V_A^{(0)}}{2}\big(1 + \rho_y\big),$$

the second term being half the $\mathrm{Cov}[e_m, g_p] \neq 0$ we found between the mates themselves, inherited by their child. A parent's *environment* is correlated with their child's genetic value; two siblings' environments are not correlated with each other's. This asymmetry between lineal and collateral relatives of the same degree is absent under random mating, and it is a signature of assortative mating rather than an artefact of it.

For user to do: Still need to revise rest of article below, agents should not edit below this point unless explicitly directed.

[AM-equilibrium] Intergenerational equilibrium

Assortment has been operating at the same strength for long enough that the variance components have stopped changing from one generation to the next.

Commonly bundled into: equilibrium assortative mating; reached after roughly six to ten generations [8].

Phenotypic scale

Assortative mating causes changes in the variance components of the trait over time. We standardize $\mathrm{Var}[y]$ to 1 in the base randomly mating population, denoted by $V_P^{(0)} = V_A^{(0)} + V_E = 1$, so that $h^2_{(0)} = V_A^{(0)}$. Quantities at equilibrium carry the superscript (eq). The heritability is always measured with variance components belonging to the same generation:

$$h^2_{(\mathrm{eq})} = \frac{V_A^{(\mathrm{eq})}}{V_P^{(\mathrm{eq})}} = \frac{V_A^{(\mathrm{eq})}}{V_A^{(\mathrm{eq})} + V_E}.$$

Mate genetic value correlation

Due to the correlation in mates' phenotypic values, their genetic values are also correlated. Per **[Primary-AM]**:

$$\begin{aligned}
\mathrm{Corr}[g_m, g_p] &= \rho_y \cdot \mathrm{Corr}[g, y]^2 \\
&= \rho_y \cdot \Big(\frac{\mathrm{Cov}[g, y]}{\sqrt{\mathrm{Var}[g] \cdot \mathrm{Var}[y]}}\Big)^2 \\
&= \rho_y \cdot \frac{\mathrm{Cov}[g, g + e]^2}{V_A^{(\mathrm{eq})} \cdot VPeq} \\
&= \rho_y \cdot \frac{(\mathrm{Cov}[g, g] + \mathrm{Cov}[g, e])^2}{V_A^{(\mathrm{eq})} \cdot V_P^{(\mathrm{eq})}} \\
&= \rho_y \cdot \frac{(\mathrm{Var}[g] + 0)^2}{V_A^{(\mathrm{eq})} \cdot V_P^{(\mathrm{eq})}} \\
&= \rho_y \cdot \frac{(V_A^{(\mathrm{eq})})^2}{V_A^{(\mathrm{eq})} \cdot V_P^{(\mathrm{eq})}} \\
&= \rho_y \cdot \frac{V_A^{(\mathrm{eq})}}{V_P^{(\mathrm{eq})}}
\end{aligned}$$

$$\boxed{\rho_g \;=\; \mathrm{Corr}[g_m, g_p] \;=\; \rho_y\, h^2_{(\mathrm{eq})}} \tag{22}$$

2.3 Assortative mating inflates genetic variance

Before looking at the covariances between individuals, we will first consider the variance of the phenotype and its components in the presence of assortative mating. First, we'll consider the distribution of an offspring's genetic value in the presence of assortative mating.

Write assumption for M_c (number of causal variants) being large

Write approximation that genetic values are normally distributed due to central limit theorem

Write assumption that environmental values are normally distributed

Consequence: phenotype value is also normally distributed. This allows us to use the properties of multivariate normal distributions in our analysis.

Partners with correlated genetic values produce offspring whose two inherited halves are correlated, which raises the variance of the offspring generation. The following seems to me like it should be written as an assumption. e.g. we are defining s_o. Furthermore, we have to justify why g_o is centered around g_m and g_f. This is trivial when talking about individual variants, but requires reusing some assumptions about uniform distribution of effect sizes across genome. So, we may need to reuse the assumptions from earlier. To see how far this goes, write the offspring's genetic value in terms of the parents',

$$g_o \;=\; \tfrac{1}{2}\,(g_m + g_f) \;+\; s_o, \tag{23}$$

where s_o is the *segregation* (or recombination) deviation: the departure of the transmitted haplotypes from the parental averages (wait this is wrong, it is the departure in genetic value, not the just the haplotypes. Once again, I think we need to be careful when working with genetic values.), which is mean-zero and uncorrelated with g_m and g_f.

I don't think we've sufficiently justified this next formula/section. I feel like it'd be better to start with the single-variant effect on increased genotype/effect*genotype variance , and then showing how it extends to the genetic value itself. Because, e.g., why would the segregation variance stay the same across generations even though other variance components are changing? Perhaps the increase is too neglible, but we need to demonstrate that first through an approximation. The variance of s_o is the one quantity in this recursion that assortment leaves alone. Segregation variance is generated by the shuffling of alleles *within* each parent, so it depends only on allele frequencies, and under **[Primary-AM]** with no selection the allele frequencies do not change. Evaluating it in the base population, where $\mathrm{Var}[g_o] = \mathrm{Var}[g_m] = V_A^{(0)}$ and $\rho_g = 0$, gives

$$V_K \;:=\; \mathrm{Var}[s_o] \;=\; V_A^{(0)} - \tfrac{1}{2}V_A^{(0)} \;=\; \tfrac{1}{2}V_A^{(0)}, \tag{24}$$

and this value carries over to every later generation. Taking the variance of (23) in a generation with additive variance $V_A(t)$ therefore gives the recursion

$$V_A(t+1) \;=\; \frac{V_A^{(0)}}{2} \;+\; V_A(t)\left(\frac{1+\rho_g(t)}{2}\right), \tag{25}$$

in agreement with Sunde et al. [8]. Setting $V_A(t+1) = V_A(t) = V_A^{(\mathrm{eq})}$ and solving,

$$\boxed{V_A^{(\mathrm{eq})} \;=\; \frac{V_A^{(0)}}{1-\rho_g}} \tag{26}$$

Equations (22) and (26) are coupled: the variance depends on ρ_g, and ρ_g depends on the heritability, which depends on the variance. Substituting one into the other and using $V_E = 1 - V_A^{(0)}$ yields a quadratic in ρ_g,

$$(1 - V_A^{(0)})\,\rho_g^2 \;-\; \rho_g \;+\; \rho_y V_A^{(0)} \;=\; 0, \qquad \rho_g \;=\; \frac{1 - \sqrt{1 - 4\rho_y V_A^{(0)}(1 - V_A^{(0)})}}{2\,(1 - V_A^{(0)})}, \tag{27}$$

taking the root that satisfies $\rho_g \to 0$ as $\rho_y \to 0$. Expanding the square root to first order gives $\rho_g \approx \rho_y V_A^{(0)} = \rho_y h^2_{(0)}$, the familiar weak-assortment approximation in which the base-population heritability may be used in place of the equilibrium one. It is only an approximation, and it understates ρ_g.

As a worked example, take a trait with $h^2_{(0)} = 0.40$ under assortment of $\rho_y = 0.30$. Then (27) gives $\rho_g = 0.130$, against the first-order value $\rho_y h^2_{(0)} = 0.120$. The additive variance rises from $V_A^{(0)} = 0.400$ to $V_A^{(\mathrm{eq})} = 0.460$, an increase of 15%, and the heritability from 0.400 to $h^2_{(\mathrm{eq})} = 0.434$.

Genic variance and gametic phase disequilibrium

Expanding $\mathrm{Var}[g_i]$ over pairs of variants and separating the diagonal from the off-diagonal,

$$V_A^{(\mathrm{eq})} \;=\; \underbrace{\sum_k \mathrm{E}\big[\beta_k^2 x_{ik}^2\big]}_{V_g} \;+\; \underbrace{\sum_k \sum_{k' \neq k} \mathrm{E}[\beta_k \beta_{k'} x_{ik} x_{ik'}]}_{D}\,. \tag{28}$$

The first term V_g is the *genic* variance: the sum of the variants' individual contributions, equivalent to the variance of a trait's genetic values when causal variants are completely independent. The second term D is the contribution of correlations *between* variants, known as gametic phase disequilibrium [9].

Ordinary LD generated by physical linkage and drift is not correlated with the the cross product of causal effects, unlike the LD generated by assortative mating. Consequently, the **[Random-effects]** assumption discussed in Section 2.6 is violated.

Since allele frequencies are unchanged, the genic variance is still the base-population value, $V_g = V_A^{(0)}$, and so by (26)

$$D \;=\; V_A^{(\mathrm{eq})} - V_A^{(0)} \;=\; \rho_g\, V_A^{(\mathrm{eq})}. \tag{29}$$

2.4 What carries over from Section 1

Before working through the levels, it is worth auditing the nine assumptions of Section 1. Three of them fail, but they fail in very different ways, and separating those ways is most of the work.

Some of what assortative mating does is real but too small to matter, and it is useful to have a distinct kind of box for such statements. An assumption answers the question "what must I believe?"; the following answers "what am I neglecting, and how big is it?".

[Same-locus-AM] Assortment does not change per-variant quantities

At any single variant, assortment leaves the picture of Section 1 intact: partners' alleles are effectively uncorrelated, the two alleles within an individual are effectively independent, and the genotype variance is still $2p_k(1-p_k)$. This preserves the per-variant identity $\mathrm{E}[x_{ik}x_{jk} \mid \mathrm{IBD}_{ij}(k)] = \frac{1}{2}\mathrm{IBD}_{ij}(k)$, and with it (8) and the definition of A_{ij}.

Why the error is negligible: Assortment correlates partners' alleles at variant k only through the trait, so the induced covariance is of order $\rho_y \beta_k^2 \sim \rho_y V_A/M$ — vanishing as the trait becomes polygenic. Summed into $A_{ij} = M^{-1}\sum_k x_{ik}x_{jk}$, the M such terms contribute $O(1/M)$ in total. The excess homozygosity that follows from the same correlation is of the same order, so **[No-inbreeding]** survives as an approximation with error $O(\rho_y V_A/M)$.

With that in hand, the audit is:

Assumption	**Status**	**Comment**
[Additivity]	holds	unchanged; the mating rule does not touch the genotype-to-phenotype map
[GE-indep]	holds	within an individual, $g_i \perp e_i$ still
[No-IGE]	holds	retained by choice; parental effects are a separate architecture
[Env-indep]	**fails**	e and g become correlated *across* individuals; see §2.4
[Random-effects]	**fails**	only the clause $\mathrm{E}[\beta_k\beta_{k'} \mid \mathbf{X}] = \mathrm{E}[\beta_k\beta_{k'}]$; see §2.6
[No-mutation]	holds	unchanged
[Non-IBD-indep]	**fails**	at the same variant, negligibly (**[Same-locus-AM]**); across variants, not negligibly
[No-inbreeding]	approximate	**[Same-locus-AM]**
[Genetic-degree]	insufficient	degree alone no longer determines the answer; see §2.8

The split in **[Non-IBD-indep]** is the single most important line in that table. Section 1 stated it as one assumption, but it was doing two jobs. Its *same-variant* job — non-IBD alleles at variant k are uncorrelated — is what makes A_{ij} track π_{ij}, and assortment violates it only negligibly. Its *across-variant* job — alleles at different variants are uncorrelated — is what made the off-diagonal of Level 0 disappear, and assortment violates that one substantially. Everything distinctive about this section follows from the second failure and not the first.

Environments become correlated across individuals

[Env-indep] and [No-IGE] were used in Section 1 to conclude, at (3), that all cross terms involving e vanish and the phenotypic cross-product reduces to the genetic one. That conclusion does not survive assortment. By [Primary-AM], partners match on y, which correlates both of its components, so

$$\mathrm{Cov}[\, e_m, g_f \,] \;=\; \rho_y \cdot \frac{V_E}{V_P^{(\mathrm{eq})}} \cdot \frac{V_A^{(\mathrm{eq})}}{V_P^{(\mathrm{eq})}} \cdot V_P^{(\mathrm{eq})} \;=\; \rho_g \, V_E \;\neq\; 0.$$

An individual's environment is correlated with their partner's genes, and their partner's genes are transmitted to their children.

The reach of this is limited, and precisely so: in a monogamous pedigree, the cross term $\mathrm{Cov}[e_a, g_b]$ is nonzero exactly when a is an **ancestor** of b (or when a and b are partners). The reason is mechanical. An individual's environment is a fresh draw, so its only route into any genome is through that individual's partner, and that partner contributes genes to b only if a lies on b's line of descent. Siblings, cousins and avuncular pairs are therefore unaffected; parent–offspring and grandparent–grandchild pairs are not. We handle this by carrying lineal pairs separately below rather than by discarding (3).

2.5 Level 0 — unchanged

Conditioning on both individuals' causal genotypes and the true effect sizes leaves nothing to average over, so (4) holds exactly as written. Level 0 is a statement about the genotype-to-phenotype map alone, and no mating rule can disturb it. That it survives every architecture untouched is a feature of where the tower starts, not a coincidence.

2.6 Level 1 — the off-diagonal no longer collapses

Section 1 moved from Level 0 to Level 1 by averaging over the effects and observing that the off-diagonal $k \neq k'$ terms vanish. Two separate clauses of [Random-effects] were needed for that: the effects are marginally uncorrelated, $\mathrm{E}[\beta_k \beta_{k'}] = 0$, and their cross-products are independent of the genotypes, $\mathrm{E}[\beta_k \beta_{k'} \mid \mathbf{X}] = \mathrm{E}[\beta_k \beta_{k'}]$.

The first clause survives assortative mating. Assortment does not reach back and change the effect sizes; it changes which genotypes are common.

The second clause does not survive, and it is worth seeing why the failure is unavoidable rather than incidental. Under assortment, the correlations that build up between variants are generated *by* the effects: it is precisely the trait-increasing alleles that end up travelling together. The observed pattern of correlation in $\mathbf{X}$ is therefore informative about which variants have effects and with what signs, so conditioning on $\mathbf{X}$ shifts our belief about $\beta_k \beta_{k'}$. Formally, $\mathrm{E}[\beta_k \beta_{k'} \mid \mathbf{X}]$ is positive for pairs of variants that are positively correlated in the sample, and the off-diagonal sum no longer disappears:

$$\mathrm{E}[\, g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j \,] \;=\; \underbrace{\sum_k \mathrm{E}[\beta_k^2]\, x_{ik} x_{jk}}_{V_A^{(0)} A_{ij}} \;+\; \underbrace{\sum_k \sum_{k' \neq k} \mathrm{E}[\beta_k \beta_{k'} \mid \mathbf{X}]\, x_{ik} x_{jk'}}_{\text{disequilibrium term}} . \tag{30}$$

The first term is the Level-1 result of Section 1, with the additive variance replaced by the *genic* variance $V_A^{(0)}$ of (28) — the genotype matrix on its own can only recover the per-variant contributions. The second term is new, and it is what the remainder of the disequilibrium D lives in.

Note the shape of (30): two relatedness-like quantities added together, not one quantity rescaled. This is the form that methods for unrelated individuals take, where the second term is estimated with a second relatedness matrix alongside the GRM [9]. We do not develop it here.

Two things to settle before writing out the disequilibrium term explicitly. First, the natural pair-level definition weights $x_{ik}x_{jk'}$ by the population LD between k and k'; the matrix $\mathbf{G}^2$ used in practice is an estimator of this that sums over *other* individuals, and so is a property of the sample rather than of the pair — which sits awkwardly with this article's framing. Second, it needs a decision about whether to restrict to between-chromosome pairs, as is done in practice to avoid confounding with LD- and MAF-dependent architecture.

2.7 The tower re-orders

At this point Section 1 descended to realized relatedness and then to genetic degree. That route is no longer available, and the reason is the screening property named at the end of Section 1.

Under random mating, π_{ij} screened off d_{ij}: once we knew what fraction of the genome a pair actually shared, their pedigree told us nothing more. Under assortative mating this is false. The disequilibrium contribution to a pair's covariance is carried by the correlations their *ancestors* had, and how many rounds of assortment separate them — information that lives in the pedigree and is not recoverable from how much genome they happen to share. Two pairs with identical π_{ij} but different pedigrees have different expected covariances.

The four information sets of Section 1 are therefore no longer a chain. Level 2 (realized relatedness) and Level 3 (pedigree) are now two *parallel* coarsenings of a common refinement — knowing both — and neither is a function of the other. We keep the level numbering, since each level still refers to the same information, but the order of derivation reverses: the pedigree is now the easy case and must be done first, the joint case follows by refinement, and realized relatedness alone is the hard case and comes last.

2.8 Level 3 — pedigree

One more generation

Everything at this level follows from a single recursion.

Lemma. *Let C be the child of P and P', and let X be any random variable such that (i) $\mathrm{Cov}[X, e_P] = 0$, and (ii) all association between X and P' runs through P's phenotype.*

Then

$$\mathrm{Cov}[\,X, g_C\,] \;=\; \frac{1+\rho_g}{2}\,\mathrm{Cov}[\,X, g_P\,]. \tag{31}$$

Proof. By (23), $\mathrm{Cov}[X, g_C] = \frac{1}{2}\mathrm{Cov}[X, g_P] + \frac{1}{2}\mathrm{Cov}[X, g_{P'}]$, the segregation term dropping out. For the second piece, condition on y_P: by (ii), $\mathrm{Cov}[X, g_{P'}] = \mathrm{Cov}\big[X, \mathrm{E}[g_{P'} \mid y_P]\big]$, and the regression of a partner's genetic value on one's own phenotype has coefficient $\mathrm{Cov}[y_P, g_{P'}]/V_P^{(\mathrm{eq})} = \rho_y V_A^{(\mathrm{eq})}/V_P^{(\mathrm{eq})} = \rho_g$ by (22). Hence $\mathrm{Cov}[X, g_{P'}] = \rho_g \mathrm{Cov}[X, y_P] = \rho_g \mathrm{Cov}[X, g_P]$, the last step using (i). $\square$

Condition (ii) is a restriction on the pedigree, and it has a familiar name.

[Monogamy] **Each individual has children with one partner**

Any two relatives are connected by a single line of descent, so a chain of association between them traverses at most one assortment link that is not on that line.

Commonly bundled into: a pedigree with no half-sibships; the leading real-world violation is an individual having children with more than one partner.

Collateral relatives

For two *collateral* relatives — individuals related through a common ancestor, neither descended from the other — Section 2.4 showed that no environmental cross terms arise, so (3) still applies and we need only the genetic covariance. The base case is a pair of full siblings, children of m and f. Expanding both siblings with (23),

$$\begin{aligned}\mathrm{Cov}[\,g_i, g_j\,] &= \tfrac{1}{4}\Big(\mathrm{Var}[g_m] + \mathrm{Var}[g_f] + 2\mathrm{Cov}[g_m, g_f]\Big)\\ &= \tfrac{1}{4}\Big(V_A^{(\mathrm{eq})} + V_A^{(\mathrm{eq})} + 2\rho_g V_A^{(\mathrm{eq})}\Big) \;=\; V_A^{(\mathrm{eq})}\,\frac{1+\rho_g}{2},\end{aligned}$$

using that the two siblings' transmissions from a given parent are conditionally independent given that parent, exactly as in Section 1. Applying (31) once for each further generation of separation gives, for a collateral pair of degree d_{ij},

$$\boxed{\mathrm{E}[\,y_i y_j \mid d_{ij}\,] \;=\; V_A^{(\mathrm{eq})}\left(\frac{1+\rho_g}{2}\right)^{d_{ij}}} \tag{32}$$

in agreement with Sunde et al. [8]. Setting $\rho_g = 0$ recovers $V_A^{(0)} 2^{-d_{ij}}$, the Level-3 result of Section 1.

Lineal relatives

For a pair in which a is an ancestor of b, the environmental cross term of Section 2.4 contributes and the base case changes. For a parent and their offspring it is easiest to work with $\mathrm{Cov}[y_m, g_o]$ directly, since e_o is a fresh draw:

$$\mathrm{Cov}[\,y_m, y_o\,] = \mathrm{Cov}[\,y_m, g_o\,] = \tfrac{1}{2}\Big(\mathrm{Cov}[y_m, g_m] + \mathrm{Cov}[y_m, g_f]\Big)$$
$$= \tfrac{1}{2}\Big(V_A^{(\mathrm{eq})} + \rho_y V_A^{(\mathrm{eq})}\Big) \;=\; V_A^{(\mathrm{eq})}\,\frac{1+\rho_y}{2}.$$

The second term is the whole story: the focal parent's *phenotype* predicts the other parent's genetic value with regression coefficient $\rho_y V_A^{(\mathrm{eq})}/V_P^{(\mathrm{eq})}$, and half of those genes reach the child. Iterating (31) for the remaining generations gives

$$\boxed{\mathrm{E}[\,y_a y_b \mid d_{ab}\,] \;=\; V_A^{(\mathrm{eq})}\left(\frac{1+\rho_y}{2}\right)\left(\frac{1+\rho_g}{2}\right)^{d_{ab}-1}} \tag{33}$$

Comparing (32) and (33), the two differ in one factor, ρ_y in place of ρ_g. Since $\rho_g = \rho_y h^2_{(\mathrm{eq})} < \rho_y$, the lineal covariance is always the larger. This is a genuine and somewhat surprising consequence: under random mating a parent–offspring pair and a full-sibling pair have identical expected covariance $V_A/2$, whereas under assortative mating the parent–offspring covariance exceeds the sibling covariance by

$$V_A^{(\mathrm{eq})}\,\frac{\rho_y - \rho_g}{2} \;=\; \frac{\rho_y V_A^{(\mathrm{eq})}\left(1 - h^2_{(\mathrm{eq})}\right)}{2}.$$

In the worked example this is a 15% excess over the sibling value. The asymmetry is purely environmental in origin — it is the parent's own environment, correlated with the other parent's genes — and a model that tracked only genetic covariances would miss it entirely.

Degree is no longer sufficient

Equations (32) and (33) disagree at the same d, which already shows that the degree alone does not determine the answer. Half-siblings make the point again, from the other direction. They are second-degree relatives, but they violate **[Monogamy]**, and the two outer parents P_1 and P_2 — both of whom assorted on the shared parent P — are themselves correlated:

$$\mathrm{Cov}[\,g_{P_1}, g_{P_2}\,] \;=\; \mathrm{Var}[y_P]\left(\frac{\rho_y V_A^{(\mathrm{eq})}}{V_P^{(\mathrm{eq})}}\right)^2 \;=\; \rho_g^2\, V_P^{(\mathrm{eq})},$$

which gives

$$\mathrm{E}[\,y_i y_j \mid \text{half-sibs}\,] \;=\; \tfrac{1}{4}\Big(V_A^{(\mathrm{eq})}(1+2\rho_g) + \rho_g^2\, V_P^{(\mathrm{eq})}\Big).$$

This exceeds the collateral value $\frac{1}{4}V_A^{(\mathrm{eq})}(1+\rho_g)^2$ by $\frac{1}{4}\rho_g^2 V_E$. So three different second-degree relationships — grandparent–grandchild, avuncular, and half-sibling — have three different expected covariances.

[Genetic-degree] must therefore be strengthened: the conditioning variable at Level 3 is not the degree but the **pedigree relationship**, of which the degree is only a summary. The same calculation also shows that individuals with no common ancestor at all can be correlated: two people who both had children with the same person have expected phenotypic cross-product $\rho_y^2\, V_P^{(\mathrm{eq})}$, and partners themselves $\rho_y V_P^{(\mathrm{eq})}$. Under random mating both are zero.

2.9 Levels 2 and 3 together — pedigree and realized relatedness

Knowing the pedigree fixes the *expected* sharing, $\bar{\pi}_d = 2^{-d}$, but two pairs with the same relationship still differ in how much genome they actually share. Section 1 handled that variation at Level 2; here we ask what it buys once the pedigree is already known. Following the natural parameterisation, we write the realized sharing as its pedigree expectation plus a deviation,

$$\pi_{ij} \;=\; \bar{\pi}_{d_{ij}} + \delta_{ij}, \qquad \mathrm{E}[\delta_{ij} \mid d_{ij}] = 0.$$

The two terms of (30) respond to δ_{ij} very differently, and that is the whole content of this level. The genic term is a sum of *same-variant* products, and by **[Same-locus-AM]** the argument of Section 1 goes through unchanged: its conditional expectation is $V_A^{(0)}\pi_{ij}$, tracking the realized sharing exactly. The disequilibrium term is a sum of *cross-variant* products, and it behaves quite differently.

[GPD-pedigree] **The disequilibrium contribution depends only on the pedigree**

The cross-variant part of a pair's covariance is a function of their pedigree relationship alone, and does not depend on the realized sharing π_{ij}.

Why the error is negligible: A cross-variant term $\mathrm{E}[x_{ik}x_{jk'}]$ with $k \neq k'$ is determined by *which* ancestral genomes the two alleles descend from — fixed by the pedigree — and not by whether i and j happen to be IBD at either locus, since the two alleles are at different loci. A residual dependence does exist, because a pair sharing more genome is also more likely to be IBD at k *and* k' jointly, but for unlinked variants this is a second-order effect: the correction is of order $\rho_g\delta_{ij}$, being the product of the disequilibrium share (29) and the sharing deviation. Deriving it exactly is left open.

Putting the two together, the expected cross-product is the pedigree value plus a term linear in the sharing deviation. The coefficient on that term is fixed by requiring consistency with Level 3: averaging over δ_{ij} must return the Level-3 value for that relationship. Hence, writing $C_{ij}(\text{pedigree})$ for the Level-3 value of the pair's relationship,

$$\boxed{\mathrm{E}[\,y_i y_j \mid \text{pedigree}, \pi_{ij}\,] \;=\; \underbrace{C_{ij}(\text{pedigree})}_{\text{from (32) or (33)}} + V_A^{(0)}\left(\pi_{ij} - \bar{\pi}_{d_{ij}}\right)} \tag{34}$$

The structure of (34) is worth dwelling on, because it separates two things that Section 1 could not tell apart. The *baseline* is set by the pedigree and is inflated by assortment. The *response to realized sharing* is not inflated at all: its coefficient is $V_A^{(0)}$, the base-population — equivalently genic — additive variance, because only the same-variant part of the covariance responds to how much genome a pair actually shares. Under random mating the two coincide, $V_A^{(0)} = V_A^{(\mathrm{eq})}$, and (34) collapses back to $V_A\pi_{ij}$.

This suggests a way to see assortative mating directly, which we note without pursuing. Within a single relationship class, regressing pairs' phenotypic cross-products on their realized relatedness gives a slope that estimates $V_A^{(0)}$ and an intercept that carries the assortment-inflated baseline. The two would agree under random mating, so the gap between them is a signature of assortment, obtainable without ever comparing across relationship classes. Whether the residual $O(\rho_g \delta_{ij})$ term of [GPD-pedigree] is small enough for this to be clean is the open question flagged in that box.

2.10 Level 2 — realized relatedness alone

The last case is the one Section 1 found easiest and assortative mating makes hardest: we observe how much genome a pair shares, but know nothing about their pedigree. Because the pedigree is exactly what carries the disequilibrium contribution, we cannot evaluate (34) directly; we have to infer a pedigree from the sharing.

The natural device is to invert the degree-to-relatedness map of Section 1. Since $\bar{\pi}_d = 2^{-d}$, a pair sharing a fraction π_{ij} of their genome is, on average, as related as a relative of degree $-\log_2 \pi_{ij}$.

[Effective-degree] Realized sharing stands in for a degree

A pair's realized relatedness is treated as though it arose from a collateral relationship of *effective degree*

$$d_{ij}^{\text{eff}} \;:=\; -\log_2 \pi_{ij},$$

which is generally not an integer.

Why the error is negligible: The substitution is exact whenever π_{ij} equals its pedigree expectation for some collateral relationship, i.e. on the grid $\pi \in \{2^{-1}, 2^{-2}, \dots\}$; off the grid it interpolates. The interpolation is benign for distant pairs, where many relationships of similar degree contribute and the sharing distribution is smooth, and poor for close pairs, where a handful of discrete relationships dominate and, as §2.8 showed, relationships of the same degree can differ. It is therefore an approximation for the distant tail, not a replacement for (34).

Substituting d_{ij}^{eff} into (32) and simplifying,

$$\left(\frac{1+\rho_g}{2}\right)^{-\log_2 \pi_{ij}} = 2^{\log_2 \pi_{ij}} \cdot (1+\rho_g)^{-\log_2 \pi_{ij}} = \pi_{ij} \cdot \pi_{ij}^{-\log_2(1+\rho_g)},$$

so that the expected cross-product is a *power law* in the realized relatedness:

$$\boxed{\mathrm{E}[\, y_i y_j \mid \pi_{ij} \,] \;=\; V_A^{(\text{eq})}\, \pi_{ij}^{\,1-\log_2(1+\rho_g)}} \tag{35}$$

Three things are worth reading off (35).

First, it contains Section 1 as a special case. With $\rho_g = 0$ the exponent is exactly 1, the relationship is linear, and we recover $\mathrm{E}[y_i y_j \mid \pi_{ij}] = V_A \pi_{ij}$. Assortative mating shows up as a single number: a deficit in the exponent.

Second, the exponent is less than one, so the curve is concave. The proportional inflation grows as pairs become more distant — in the worked example the exponent is 0.823, and while sibling covariance is inflated by 30% relative to a randomly mating population with the same base heritability, first-cousin covariance is inflated by 66%. This is the reason assortative mating is more visible in distant relatives than in close ones, and the reason it inflates estimates that lean on distant or nominally unrelated pairs.

Third, (35) is linear on a log–log scale,

$$\log \mathrm{E}[\, y_i y_j \mid \pi_{ij}\,] \;=\; \log V_A^{(\mathrm{eq})} \;+\; \bigl(1 - \log_2(1+\rho_g)\bigr)\, \log \pi_{ij},$$

so the slope of expected cross-product against realized relatedness, both on log scales, estimates ρ_g through its shortfall below one. Combined with the within-class result of §2.9, this gives two distinguishable signatures: *within* a relationship class the response to sharing is linear with slope $V_A^{(0)}$, while *across* classes it is a power law with exponent below one. Under random mating these are the same line. The wedge between them is assortative mating.

Finally, note where (35) points as $\pi_{ij} \to 0$. The covariance vanishes, but more slowly than linearly, and the pairs contributing in that limit are the nominally unrelated ones. This is the regime in which the additive form (30) is the natural description, and where methods that fit a second, disequilibrium relatedness matrix alongside the GRM operate [9].

2.11 Summary

The same four information sets as Section 1, but no longer arranged in a chain. Level 2 and Level 3 are parallel coarsenings of their common refinement, and the derivation runs $0 \to 1 \to 3 \to (2,3) \to 2$ rather than top to bottom.

Level	**Conditioned on**	$\mathrm{E}[y_i y_j \mid \cdot]$	**New assumptions**
0	causal genotypes, effects	$g_i g_j$	— (unchanged)
1	genotypes $\mathbf{x}_i, \mathbf{x}_j$	$V_A^{(0)} A_{ij}$ + disequilibrium	**[Random-effects]** weakened
3	pedigree	$V_A^{(\mathrm{eq})} \left(\frac{1+\rho_g}{2}\right)^d$ (collateral)	**[Pheno-AM]**, **[Primary-AM]**, **[AM-equilibrium]**, **[Monogamy]**
$(2,3)$	pedigree and π_{ij}	$C_{ij} + V_A^{(0)}(\pi_{ij} - \bar{\pi}_d)$	**[Same-locus-AM]**, **[GPD-pedigree]**
2	π_{ij}	$V_A^{(\mathrm{eq})}\, \pi_{ij}^{\,1-\log_2(1+\rho_g)}$	**[Effective-degree]**

Reduced to one sentence: assortative mating adds a second channel of resemblance that travels through the pedigree rather than through shared segments, so relatedness and genealogy stop being interchangeable. Everything else in this section is bookkeeping for that one fact.

Still to do in this section: the explicit disequilibrium relatedness matrix at Level 1 and its estimation in samples of unrelated individuals; the exact size of the residual term in **[GPD-pedigree]**; and the non-equilibrium case, where a mismatch between observed and predicted correlations dates the onset of assortment [8].

References

[1] Sewall Wright. Correlation and causation. *Journal of Agricultural Research*, 20:557–585, 1921.

[2] Sewall Wright. The method of path coefficients. *The Annals of Mathematical Statistics*, 5(3):161–215, 1934. URL `http://www.jstor.org/stable/2957502`.

[3] Sewall Wright. Systems of mating. I. the biometric relations between parent and offspring. *Genetics*, 6(2):111–123, 1921. doi: 10.1093/genetics/6.2.111.

[4] Sewall Wright. Systems of mating. III. assortative mating based on somatic resemblance. *Genetics*, 6(2):144–161, 1921. doi: 10.1093/genetics/6.2.144.

[5] R A Fisher. The correlation between relatives on the supposition of mendelian inheritance. *Transactions of the Royal Society of Edinburgh*, 52:399–433, 1918. doi: 10.1017/s0080456800012163.

[6] Hans Fredrik Sunde, Espen Moen Eilertsen, and Fartein Ask Torvik. Understanding indirect assortative mating and its intergenerational consequences for educational attainment. *Nat. Commun.*, 16(1):5264, 6 June 2025.

[7] Loic Yengo, Matthew R Robinson, Matthew C Keller, Kathryn E Kemper, Yuanhao Yang, Maciej Trzaskowski, Jacob Gratten, Patrick Turley, David Cesarini, Daniel J Benjamin, Naomi R Wray, Michael E Goddard, Jian Yang, and Peter M Visscher. Imprint of assortative mating on the human genome. *Nature Human Behaviour*, 2: 948–954, 2018. doi: 10.1038/s41562-018-0476-3.

[8] Hans Fredrik Sunde, Nikolai Haahjem Eftedal, Rosa Cheesman, Elizabeth C Corfield, Thomas H Kleppestø, Anne Caroline Seierstad, Eivind Ystrom, Espen Moen Eilertsen, and Fartein Ask Torvik. Genetic similarity between relatives provides evidence on the presence and history of assortative mating. *Nat. Commun.*, 15(1):2641, 2024. doi: 10.1038/s41467-024-46939-9.

[9] Yuanxiang Zhang, Saori Sakaue, Sam Morris, Mariaelisa Graff, Kenichi Yamamoto, Zhengming Chen, Liming Li, Matthew C Keller, Michael E Goddard, Robin G Walters, Yukinori Okada, Peter M Visscher, and Loic Yengo. The contribution of gametic phase disequilibrium to the heritability of complex traits. *Nat. Genet.*, 2025. doi: 10.1038/s41588-025-02192-4.